Scaling LLM-Based Knowledge Graph Generation: A Case Study of Italian Geopolitical News

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***Abstract*—Geopolitical news provides vast amounts of information essential for understanding international relations and political events. However, organizing this information into a coherent, structured format poses challenges due to the complexity and dynamic nature of the domain. This paper introduces a scalable system leveraging Large Language Models to build continuously updated Knowledge Graphs from Italian geopolitical news. The system features a modular architecture, including a Collector Node for scalable article extraction, a Redis-based reliable queue to manage large-scale data ingestion, and a Named Entity Recognition/Relation Extraction Engine to standardize entity-relation triples. The framework addresses key challenges, such as continuous updating and hallucination mitigation, ensuring the reliability of the graph. Our evaluations demonstrate significant improvements in scalability, uniformity of extracted triples, and graph accuracy, making this architecture particularly suitable for real-time geopolitical analysis.**

***Index Terms*—Knowledge Graph (KG), Retrieval-Augmented Generation (RAG), Named Entity Recognition (NER), Relation Extraction (RE), Large Language Models (LLMs).**

I. Introduction

Knowledge Graphs (KGs) have become essential tools for representing structured knowledge and have been widely applied in tasks such as question answering [1], recommender systems [2], and web search [3]. Traditional methods for constructing and maintaining KGs typically involve multi-stage processes, including entity discovery and relation extraction. However, these processes are inherently complex, often resulting in incompleteness due to the difficulty of capturing all possible entities and relationships from vast and diverse data sources [4]. Additionally, traditional KGs often suffer from insufficient integration of textual information, as they rely heavily on structured or semi-structured data, limiting their ability to capture context-rich information from unstructured text [5]. Scalability also presents a significant challenge: as data volume grows, conventional KG construction methods require manual intervention to maintain updates [6]. These limitations hinder the adaptability and effectiveness of traditional KGs, particularly in dynamic domains requiring continuous data integration, such as news analysis.

To overcome these limitations, we build upon recent advancements in Large Language Models (LLMs), which enable the augmentation of KGs with rich contextual information. We present a novel system architecture specifically tailored for constructing and maintaining a KG from extensive volumes of news articles, utilizing an LLM-driven pipeline. The system employs a modular architecture that integrates several components: a Collector Node for systematic article retrieval, leveraging a reliable queuing mechanism that prevents data loss by buffering articles and enabling recovery in case of processing failures. For the extraction and structuring of entity-relation triples, we implement an LLM-based Named Entity Recognition and Relation Extraction (NER/RE) Engine that combines LLM capabilities with rule-based filters, leveraging the advanced contextual understanding of LLMs to capture nuanced relationships from unstructured text and ensuring that only valid triples are integrated into the KG. The modular and horizontally scalable design allows for real-time ingestion and continuous updates, making the system particularly suited for geopolitical news analysis, where information flows are constant and diverse. To demonstrate the effectiveness of the proposed system, we apply it to a case study involving the analysis of Italian geopolitical news articles sourced from *notiziegeopolitiche.net*[1]. The resulting KG is continuously updated, capturing real-time developments in Italian and international geopolitical contexts.

II. Related Work

Recent research has explored integrating LLMs to augment KGs, as highlighted by Pan et al. [6]. LLMs have been applied to various KG-related tasks such as embedding, completion, construction, and KG-to-text generation, demonstrating their ability to handle both structured and unstructured data while improving performance in these areas. End-to-end KG construction techniques, such as those by Zhu et al. [7], leverage LLMs to automate the construction process. For example, Wei et al. [8] utilize ChatGPT for NER and RE, effectively streamlining KG creation from raw text, while Han et al. [9] introduce a prompting framework to correct errors in LLM outputs. Additionally, Bosselut et al. [10] explore distilling

[1]https://www.notiziegeopolitiche.net/

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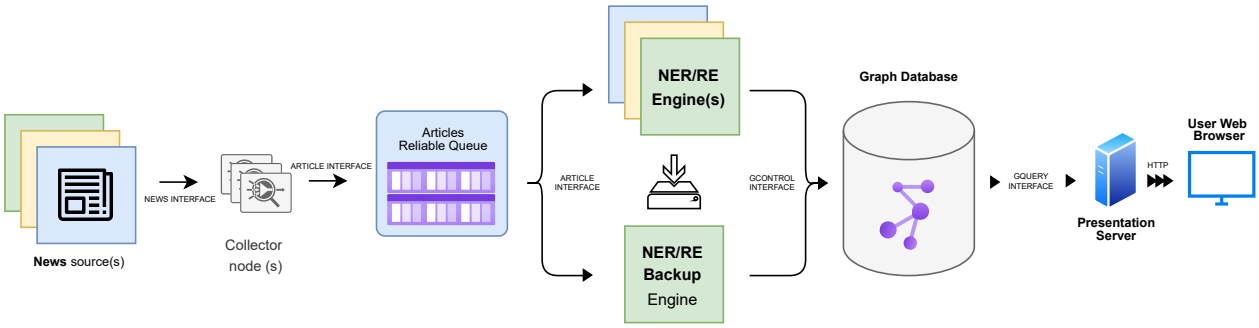

Fig. 1. The pipeline illustrates the workflow for processing articles from different news sources. The **Collector node** gathers articles and standardizes them, storing them in the **Articles Reliable Queue** to ensure process reliability. The **NER/RE Engine** extracts entities and relationships from the articles, saving them into a **Graph Database**. In case of failures, the **NER/RE Backup Engine** ensures data recovery, maintaining system scalability. Finally, a **Presentation Server** provides an interactive dashboard for visualizing the results and performing in-depth analysis on the graph.

commonsense knowledge from LLMs to construct high-quality KGs using a seed set of existing tuples.

Despite these advancements in the above-mentioned works, most LLM-driven KG construction studies are limited to domains unrelated to geopolitics. Furthermore, challenges such as hallucination in LLM outputs remain significant obstacles to reliable KG construction. Finally, current research predominantly focuses on English-language data, with limited attention to multilingual sources [11], which restricts the applicability of these models in globally diverse contexts. Our proposed system addresses these limitations by incorporating advanced queue management, scalable article ingestion, and dynamic triple extraction, positioning our work as an innovative solution to real-time KG construction in the geopolitical domain.

III. Methodology

The system architecture, depicted in Figure 1, integrates several components into an advanced pipeline designed for the efficient and continuous ingestion, processing and extraction of entity-relation triples from large volumes of news articles, ultimately constructing a comprehensive KG. The architecture adopts a modular, horizontally scalable design to ensure uninterrupted and reliable processing, even as data volumes increase.

A. Collector Node

The Collector Node module is responsible for extracting raw articles through the `NEWS INTERFACE`, an adaptable component designed to accommodate a variety of sources, enabling the system to efficiently ingest data from multiple types of providers without additional customization. The system retrieves articles in manageable batches to optimize memory usage and processing efficiency. Subsequently, the articles are standardized according to a predefined structure via the `ARTICLE INTERFACE`:

Document Structure

ARTICLE:

- **Title:** [Article's title]
- **Link:** [Article's URL]
- **Date:** [Publication date]
- **Text:** [Full article text]

Multiple instances of the Collector Node can be deployed in parallel, thereby enhancing throughput and enabling horizontal scaling to process large volumes of articles simultaneously from diverse data sources. This parallelization ensures efficient and timely data ingestion, which is essential for real-time or near-real-time geopolitical analysis.

B. Articles Reliable Queue

The Articles Reliable Queue is a specialized module for efficient article processing. It implements a dual-queue system with a main queue and a backup queue. Upon ingestion, articles are placed in the main queue for batch processing. Before the processing begins, the articles are temporarily copied to the backup queue until successful processing is confirmed. This dual-queue architecture significantly enhances availability and fault tolerance. In the event of a processing failure, the system can access the backup queue to reprocess articles, preventing data loss and ensuring pipeline integrity. Additionally, this architecture decouples article ingestion from processing, enabling each stage to operate independently and asynchronously. This separation allows for continuous article ingestion, even when processing requires more computational resources.

C. NER/RE Engine

The NER/RE Engine performs the core analysis of the system: extracting structured information from raw text. The engine processes batches of articles retrieved from the Articles Reliable Queue via `ARTICLE INTERFACE`. The entity-relation extraction pipeline consists of a hybrid approach that incorporates an LLM alongside rule-based filters. The

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LLM identifies and classifies entities (e.g., people, organizations, locations) and the relationships between them, while deterministic filters mitigate hallucination by ensuring that only valid triples `(entity1[category1] - relation - entity2[category2])` are produced according to the Entity-Relation Model. This model specifies the entity types and relationships that the system can identify, serving as the foundation for its analytical capabilities and dictating the structure of the resulting KG. Table I details the entity types considered in our case study.

TABLE I
ENTITY TYPES

Entity Type	Example
Person	Joe Biden, Angela Merkel, Elon Musk
Organization	United Nations, World Bank, Google
Location	Italy, New York, Asia
Date or Time Period	March 12, 2024, second quarter, 1990s
Measurement or Quantity	100 million dollars, 5%, 200 billion cubic meters
Product or Service	iPhone, solar energy, banking services
Event	Brexit, G20, World Economic Forum
Law, Regulation, Political Program	Paris Agreement, GDPR Regulation, tax reform
Economic Sector	Technology sector, manufacturing industry
Natural Resource	Oil, natural gas, rare minerals
Government Entity	Ministry of Finance, European Central Bank
Infrastructure	Suez Canal, Nord Stream Pipeline
Technology	Artificial intelligence, blockchain, 5G
Incident or Crisis	COVID-19 pandemic, SolarWinds cyber attack

The Entity-Relation Model differentiates between two categories of relationships: Action-based (Table II) and Affective (Table III). Action-based relationships denote factual connections between entities, such as ”meeting”, ”discussing” and ”attending”, while Affective relationships express positive or negative sentiments, including ”appreciating”, ”supporting” or their opposites like ”disapproving”. Affective relationships are particularly pertinent in geopolitical and economic analyses, where public and political sentiment significantly influences the interpretation of events and decisions.

TABLE II
ACTION-BASED RELATIONSHIPS

Relationship Type	Example
Meet	Joe Biden meets Angela Merkel in Berlin.
Discuss	World leaders discuss climate issues at the G20.
Attend	Elon Musk attends a conference on technological innovation.
Announce	The World Bank announces new funding for Africa.
Report	The United Nations reports on humanitarian conditions in Syria.
Predict	Economists predict GDP growth for 2025.
Collaborate	Google and Microsoft collaborate on artificial intelligence projects.
Publish	The UN publishes a report on climate change.
Record	Italy records an increase in exports in the first quarter.

Scalability is a key feature of the NER/RE Engine, as multiple instances of the engine can be instantiated in parallel to handle large volumes of data coming from the Articles Reliable Queue. This horizontal scalability ensures that the system can maintain a high processing throughput even as the number of articles ingested increases significantly. However, it is important to note that utilizing multiple LLMs may introduce bottlenecks due to the increased computational demands. The extracted triples will be inserted into the graph database via the `GCONTROL INTERFACE`, a list of functions designed specifically for this purpose.

TABLE III
AFFECTIVE RELATIONSHIPS

Relationship Type	Positive / Negative Example
Appreciate/Disapprove	The public appreciates Joe Biden’s speeches. The public disapproves of austerity policies.
Support/Obstruct	Germany supports the EU initiatives. Germany obstructs new tax regulations.
Promote/Boycott	Google promotes sustainable technologies. Google boycotts new privacy laws.
Invest/Cut	The World Bank invests in development projects. The World Bank cuts funds for European projects.
Celebrate/Ignore	Italy celebrates Republic Day on June 2. Italy ignores the EU recommendations.
Welcome/Reject	Europe welcomes refugees. Europe rejects asylum requests.
Value/Neglect	Egypt values the Suez Canal as key infrastructure. Egypt neglects local road maintenance.
Adopt/Refuse	Companies adopt artificial intelligence. Companies refuse to implement new technologies.
Encourage/Dissuade	The government encourages technological innovation. The government dissuades foreign investment.

D. NER/RE Backup Engine

The NER/RE Backup Engine mirrors the functionality of the primary NER/RE Engine but processes articles from the backup queue. It serves as a redundancy mechanism, ensuring system resilience by reprocessing articles in case of primary engine failure. Unlike the primary engine, the backup cannot be parallelized, as its role is dedicated to fault recovery.

E. Presentation Server

The Presentation Server provides an interactive dashboard interface divided into several key sections enriched with information extracted from the Graph Database using the `GQUERY INTERFACE`. The *Homepage* displays general Knowledge Graph statistics, including the most cited entities in the selected period. The *Map Explorer* allows analysis of geographical mentions of each country in the analysed articles, with both tabular and bar graph display options. The *Trend Viewer* offers a graphical representation of the temporal trend of entity mentions, while the *Political Compass* allows one to examine the sentimental relationships between a root entity and other entities, quantifying the polarity (positive or negative) of these relationships. The *Graph Explorer* facilitates the dynamic search and visualisation of entities and their relationships based on user-selected temporal data.

IV. EXPERIMENTS

A. Experimental Setup

The pipeline was primarily executed on Google Colab, leveraging GPU resources to expedite data processing and model inference. Neo4J[2] served as the graph database, efficiently modeling entities and relationships extracted from geopolitical articles through its optimized handling of complex, interconnected data. Redis[3] functioned as the backbone of the Articles Reliable Queue, providing low-latency, in-memory queuing to support scalable, real-time ingestion for NER/RE processing. The LLM used was LLaMA-3-8B [12],

[2]https://neo4j.com/
[3]https://redis.io/

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enabling entity-relationship triple extraction. This integration of Redis and Neo4J ensured a scalable and resilient architecture, ideally suited for efficiently processing large volumes of news data.

B. Dataset

The dataset used for this study consists of geopolitical and economic articles written in Italian, sourced from the website *notiziegeopolitiche.net*, a prominent Italian outlet for political and economic reporting. These articles provide a rich collection of information, covering various geopolitical events, economic trends, and international relations.

C. Evaluation

1) Quantitative Evaluation: We tested the integration of our pipeline comparing it with conventional machine learning models for predicting relationships between entities within the graph. Utilizing GraphX and SparkML, we extracted structural graph features (PageRank, Triangle Counting, and Label Propagation) to train a Random Forest classifier. Configured with 10 trees, the Random Forest model achieved a test error rate of 0.46, outperforming baseline models such as Decision Tree and Logistic Regression, which showed error rates between 0.50 and 0.60. Other models, such as k-Nearest Neighbors and Naive Bayes, presented additional limitations, including high computational time and challenges in handling correlated data. In comparison, our pipeline, exploiting LLMs, achieved an error rate of 0.38, proving to be the most effective solution. Its ability to capture linguistic variation and identify relationships between entities significantly increases predictive accuracy compared to traditional models. We chose the error rate as the main metric to assess the accuracy of relationship prediction, thus ensuring the quality and consistency of the KG.

2) Qualitative evaluation: We tested the system's usability and provided valuable feedback. We highlighted its ability to deliver timely and accurate updates, praising both the intuitive interface and the quality of entity-relation extraction. For example, the system accurately identified relations such as `Joe Biden - Meet - Angela Merkel` and `The World Bank - Announce - New funding for Africa`, both of which were valuable for analysts tracking diplomatic engagements and financial investments. However occasional inaccuracies were observed. For instance, the relation `The World Bank - Predict - Cuts in European funding` as potentially misleading; in this case, "Predict" did not accurately reflect the entity's actual role, underscoring the need for further tuning, particularly for nuanced or ambiguous contexts.

V. Conclusion and Future Direction

This paper presents a scalable system for building Knowledge Graphs using LLMs to process Italian geopolitical news articles. The system excels in continuously updating the graph while maintaining triple uniformity and mitigating LLM hallucinations. While the system has shown promising results in real-time geopolitical analysis, future work will focus on integrating the system into larger GraphRAG workflows and extending its capabilities to cover a broader range of topics. Potential future research includes improving the system's accuracy in other languages, incorporating more advanced machine learning algorithms, and conducting large-scale deployments to further validate its scalability and effectiveness.

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